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Data Mining I

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Association Rules

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Association Rules

- **Association Rules:**
 - Aims at the extraction of compact patterns that describe subsets of data
 - Events that occur together (market basket analysis);
 - The main purpose is to establish relationships between fields;
 - Are rules of the form if X then Y;
 - Association rules provide information about things that tend to happen together.

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Association Rules

- **Association Rules:**
 - Table with a set of purchases from a supermarket, with 5 purchases and 5 items.

Client	Items
1	Orange juice, soda
2	Milk, Orange juice, Glass cleaner
3	Orange juice
4	Orange juice, detergent, soda
5	Glass cleaner, soda

- With these data we can create a table of co-occurrences with the number of times that any pair of products was purchased together.

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Association Rules

- Association Rules:**

	Orange Juice	Glass Cleaner	Milk	Soda	Detergent
1	1	0	0	1	0
2	1	1	1	0	0
3	1	0	0	0	0
4	1	0	0	1	1
5	0	1	0	1	0

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Association Rules

	Orange Juice	Glass Cleaner	Milk	Soda	Detergent
Orange Juice	4	1	1	2	1
Glass Cleaner	1	2	1	1	0
Milk	1	1	1	0	0
Soda	2	1	0	3	1
Detergent	1	0	0	1	1

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Association Rules

- From the analysis, one can conclude that:
 - Orange juice and soda are more likely purchased together than any other two items;
 - Detergent is never purchased with milk or glass cleaner;
 - Milk is never purchased with soda or detergent.

	OJ	GC	Milk	Soda	Det.
OJ	4	1	1	2	1
GC	1	2	1	1	0
Milk	1	1	1	0	0
Soda	2	1	0	3	1
Det.	1	0	0	1	1

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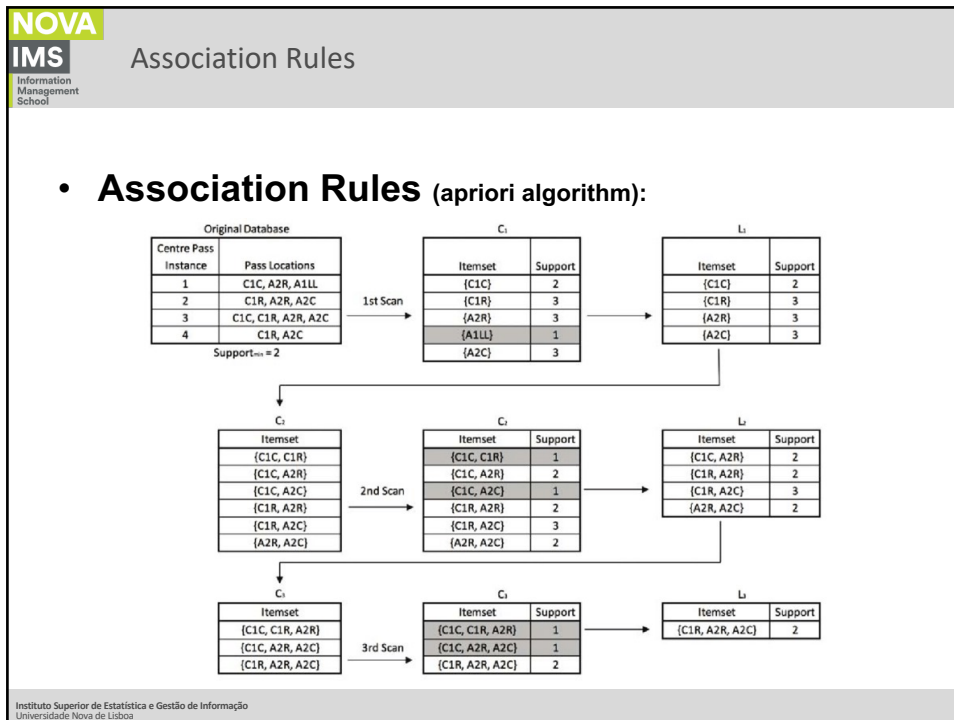
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Association Rules

- Association Rules (apriori algorithm):**
 - Frequent Itemset Generation:**
 - The algorithm starts by scanning the dataset to count the frequency of each item. It creates a list of itemsets (sets of items) that are frequent, meaning they occur in transactions more often than a predefined minimum support threshold. These frequent itemsets are called "L1" itemsets.
 - Joining to Create Candidate Itemsets:**
 - The algorithm then generates candidate itemsets for the next level (L2) by joining pairs of L1 itemsets. For example, if "milk" and "bread" were frequent individually (L1), the algorithm combines them to create a candidate itemset (milk, bread). These candidate itemsets are then checked to see if their subsets (e.g., {milk} and {bread}) are frequent.
 - Pruning Infrequent Itemsets:**
 - Candidate itemsets that have subsets that are not frequent are eliminated because, if a subset is infrequent, the superitemset cannot be frequent either. This reduces the number of itemsets to be checked.
 - Repeat the Process:**
 - Steps 2 and 3 are repeated iteratively to create candidate itemsets of higher levels (L3, L4, etc.) by joining frequent itemsets from the previous level. The algorithm continues until no more frequent itemsets can be generated, or until a certain level of itemset size is reached.
 - Association Rule Generation:**
 - Once all frequent itemsets are found, the algorithm generates association rules from these itemsets.
 - Pruning Rules:**
 - The algorithm may also prune rules that don't meet a minimum confidence threshold.

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Association Rules

- Association Rules:**
 - The rules are expressed in the form of:

“if the item A is part of an event, then the item B will also be part of the event X percent of the time”
 - The rules should not be interpreted as a direct causation, but only as an association;
 - It is not legitimate to infer rules of causality.

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Association Rules

- **How association rules work:**

*If a customer buys **shoes**, then 10% of the time also buys **socks***

*If a customer buys **plastic paint**, then 85% of the time also buys **brushes***

 - All rules have an antecedent and a consequent

*Shoes and plastic paint – **antecedents***

*Socks and brushes – **consequents***

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Association Rules

- **Venn Diagram:**

Antecedent

Consequent

Database

Combination A and C

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Association Rules

- **How association rules work:**
 - *Evaluating the quality of Association Rules:*

Confidence

Support

(Consequent Support)

Lift
 - *The most important measures to assess the rules*

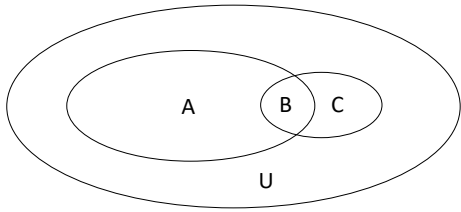
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Association Rules

- **How association rules work:**
 - *Evaluating the quality of Association Rules:*
 - **Support** – shows **how frequently** the combination occurs in the database.



$Support = \frac{B}{U}$

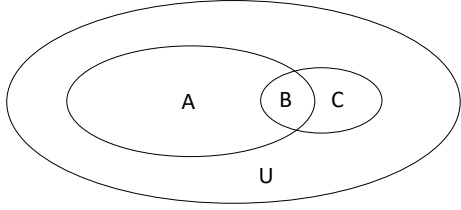
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Association Rules

- **How association rules work:**
 - *Evaluating the quality of Association Rules:*
 - **Confidence** – the **strenght of an association**, the percentage of a consequent appears given that the antecedent has occurred



$$\text{Confidence} = \frac{B}{A}$$

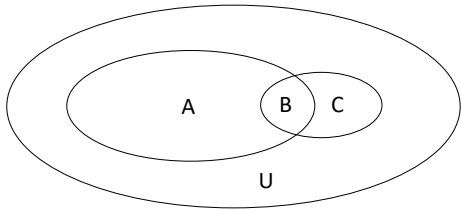
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Association Rules

- **How association rules work:**
 - *Evaluating the quality of Association Rules:*
 - **Consequent support** (expected confidence of the c) – equal to the number of consequent transactions, divided by the total number of transactions



$$\text{Consequent Support} = \frac{C}{U}$$

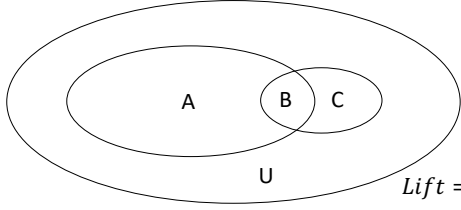
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Association Rules

- **How association rules work:**
 - *Evaluating the quality of Association Rules:*
 - **Lift** – equal to the confidence factor divided by the expected confidence. Lift is a factor by which the **likelihood** of consequent increases given an antecedent



$$\text{Lift} = \text{Confidence} / \text{Consequente Support}$$

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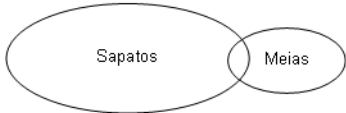
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Tabela de Transações

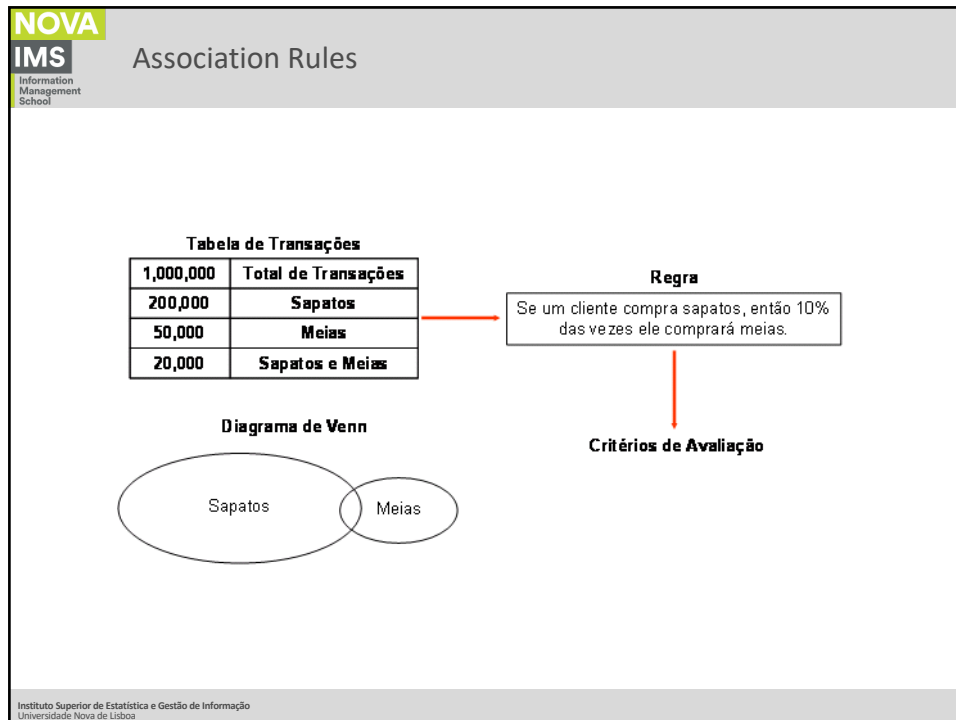
1,000,000	Total de Transações
200,000	Sapatos
50,000	Meias
20,000	Sapatos e Meias

Diagrama de Venn

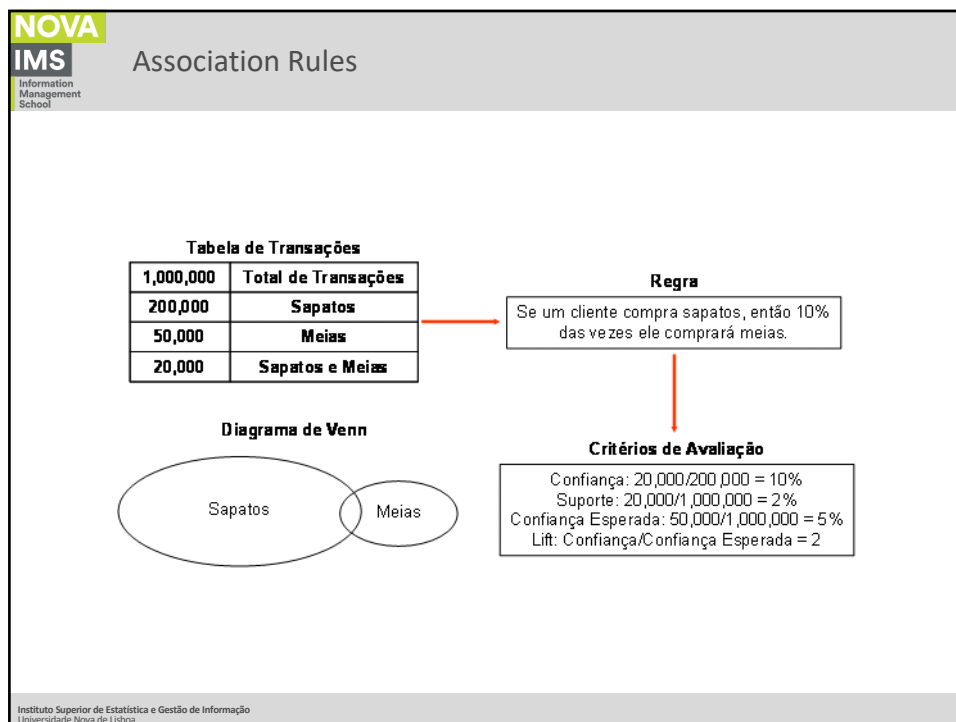


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Association Rules

- **How association rules work:**
 - *Evaluating the quality of Association Rules:*
 - A lift value of 2 indicates that **the two items (A and B) occur together twice as often as would be expected if their occurrences were independent of each other.**
 - Lift > 1: If the lift is greater than 1, it indicates a **positive association** between the two items.
 - A lift of 2, means that the two items are twice as likely to be purchased together as would be expected if their occurrences were independent. This is a strong indicator of association, and it suggests that there may be a meaningful relationship between the two items, making them good candidates for bundling or recommendations.

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Association Rules

- **How association rules work:**
 - *Evaluating the quality of Association Rules:*
 - Lift < 1 - indicates that the two items are **less likely to be bought together** than if their occurrences were independent. In other words, they have a negative association, and the presence of one item may actually reduce the likelihood of the other item being purchased.
 - Lift = 1: If the lift is equal to 1, it suggests that the **two items are independent of each other.** Their occurrence together is no more or less likely than what would be expected by chance.

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Association Rules

- **How association rules work:**
 - *Evaluating the quality of Association Rules:*
- A **credible rule** must have a good confidence factor, a high level of support and lift higher than 1
- Rules with a **high level of confidence** but with a **low support** should be interpreted with caution, as it can result in **idiosyncrasies** that arise due to the small number of cases to support the rule

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Association Rules

- **Use:**

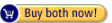
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- **Use:**

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Association Rules

- **Types of rules**
 - Trivial rules;
 - “Customers who purchase maintenance agreements are very likely to purchase large appliances.”
 - Inexplicable Rules;
 - “When a new hardware store opens, one of the most commonly sold items is toilet bowl cleaners.”
 - Actionable Rules;
 - “Wal-Mart customers who purchase Barbie dolls have a 60 percent likelihood of also purchasing one of three types of candy bars.”
 - Rules that result of promotions made.

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Association Rules

- Additional aspects (product hierarchies)**

more general

Partial Product Taxonomy

more detailed

Brands, sizes, and stock keeping units (SKUs)

Source: Berry and Linhof Data Mining Techniques

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Association Rules

- Additional aspects**
 - Available data are essential to success;
 - It takes large amounts of data;
 - Use of virtual items;
 - Allow you to get information from data beyond the information expressed in the products;
 - May include information about the purchase (paid in cash, credit card...), about the customer (new, old...) day of the week, hour of the day.

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